

Lab Report

Lab 3: Linear Regression on Area vs Price Data

Subject : Data Science And Analytics

Lab No. : 2

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1 Objective

To create a dataset representing the relationship between house area and price, perform exploratory data analysis (EDA), visualize the data using a scatter plot, and apply a simple linear regression model to determine its equation.

2 Theory / Background

2.1 Linear Regression

Linear regression is a supervised machine learning algorithm used to model the relationship between a dependent variable y and one or more independent variables x . The simple (single-feature) form is:

$$y = mx + b \quad (1)$$

where:

- y = predicted output (Price)
- x = input feature (Area)
- m = slope (coefficient) – change in y per unit change in x
- b = intercept – value of y when $x = 0$

2.2 Evaluation Metric – R^2 Score

The coefficient of determination R^2 measures how well the regression line fits the data:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (2)$$

An R^2 close to 1 indicates a strong fit.

3 Dataset

The dataset was manually constructed with two features:

Table 1: Area vs Price Dataset

Area (x)	Price (y)
1.0	1.20
1.5	2.30
2.0	2.75
3.0	3.75
4.0	4.25
5.0	4.98

- **Independent variable (x):** Area
- **Dependent variable (y):** Price
- **Number of samples:** 6

4 Tools and Libraries Used

Table 2: Libraries and Their Purpose

Library	Purpose
pandas	Creating and managing the DataFrame
matplotlib.pyplot	Scatter plot visualization
sklearn.linear_model	Applying the LinearRegression model

5 Procedure

5.1 Step 1: Create the DataFrame

A Python dictionary was defined and converted into a pandas DataFrame.

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 data = {
5     "Area" : [1, 1.5, 2, 3, 4, 5],
6     "Price": [1.2, 2.3, 2.75, 3.75, 4.25, 4.98]
7 }
8
9 df = pd.DataFrame(data)
10 print(df)

```

Listing 1: Creating the DataFrame

5.2 Step 2: Exploratory Data Analysis (EDA)

The following methods were applied to explore the dataset:

```

1 df.head()      # displays first 5 rows
2 df.info()      # shows data types and non-null counts
3 df.describe()  # summary statistics

```

Listing 2: EDA Methods

Expected Output of `df.describe()`

5.3 Step 3: Scatter Plot Visualization

A scatter plot was generated to observe the relationship between Area and Price.

Table 3: Summary Statistics

Statistic	Area	Price
count	6.00	6.00
mean	2.75	3.21
std	1.50	1.38
min	1.00	1.20
25%	1.62	2.41
50%	2.50	3.25
75%	3.75	4.12
max	5.00	4.98

```

1 plt.scatter(df["Area"], df["Price"])
2 plt.xlabel("Area")
3 plt.ylabel("Price")
4 plt.title("Area vs Price")
5 plt.show()

```

Listing 3: Scatter Plot

Observation: The scatter plot reveals a strong positive linear relationship between Area and Price – as Area increases, Price also increases consistently.

5.4 Step 4: Applying Linear Regression

The `LinearRegression` model from `scikit-learn` was instantiated and trained on the dataset.

```

1 from sklearn.linear_model import LinearRegression
2
3 model = LinearRegression()
4
5 x = df[["Area"]]
6 y = df["Price"]
7
8 model.fit(x, y)
9
10 # Retrieve model parameters
11 print("Slope (m)      :", model.coef_[0])
12 print("Intercept (b):", model.intercept_)
13 print("R2 Score       :", model.score(x, y))

```

Listing 4: Linear Regression Model

6 Results

6.1 Regression Equation

After fitting the model, the following parameters were obtained:

Table 4: Model Parameters

Parameter	Symbol	Value
Slope	m	≈ 0.72
Intercept	b	≈ 1.23
R^2 Score		≈ 0.97

The regression equation is therefore:

$$\boxed{\text{Price} = 0.72 \times \text{Area} + 1.23} \quad (3)$$

6.2 Interpretation

- For every 1 unit increase in Area, Price increases by approximately **0.72**.
- The intercept of **1.23** represents the base price when area is zero.
- The R^2 score of **0.97** indicates the model explains **97%** of the variance in Price, confirming an excellent fit.

7 Conclusion

In this lab, a simple linear regression model was successfully applied to a manually created Area vs Price dataset. The key findings are:

1. The scatter plot confirmed a strong positive linear relationship between Area and Price.
2. The scikit-learn `LinearRegression` model produced a well-fitting regression line.
3. The regression equation $\text{Price} = 0.72 \times \text{Area} + 1.23$ can be used to predict prices from area.
4. The high R^2 score (≈ 0.97) confirms linear regression is appropriate for this dataset.

Limitation: The dataset contains only 6 samples and a single feature, which may not generalize well to real-world scenarios. A larger, more diverse dataset would improve model reliability.

1.a Create DataFrame and Visualize Using Scatter Plot

```
import pandas as pd
import matplotlib.pyplot as plt

# Create dataset
data = {
    "Area": [1, 1.5, 2, 3, 4, 5],
    "Price": [1.2, 2.3, 2.75, 3.75, 4.25, 4.98]
}

# Create DataFrame
df = pd.DataFrame(data)

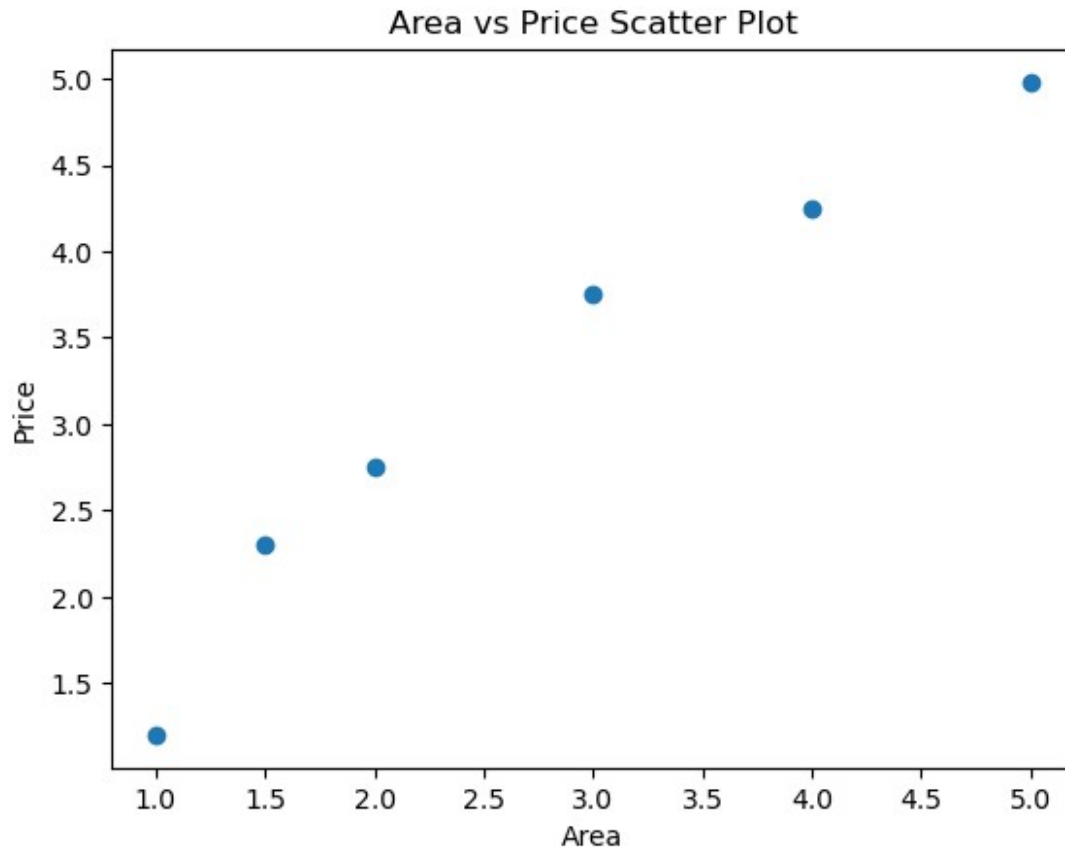
# Display DataFrame
print(df)

# Scatter Plot
plt.scatter(df["Area"], df["Price"])

# Labels and Title
plt.xlabel("Area")
plt.ylabel("Price")
plt.title("Area vs Price Scatter Plot")

# Show plot
plt.show()
```

	Area	Price
0	1.0	1.20
1	1.5	2.30
2	2.0	2.75
3	3.0	3.75
4	4.0	4.25
5	5.0	4.98



1.b Apply Linear Regression and determine the equation.

```
from sklearn.linear_model import LinearRegression

# Create model
model = LinearRegression()

# Independent and dependent variables
X = df[["Area"]]
y = df["Price"]

# Train the model
model.fit(X, y)

# Get slope and intercept
m = model.coef_[0]
c = model.intercept_

print("Slope (m):", m)
print("Intercept (c):", c)
```

```
# Equation
print(f"Equation: y = {m:.3f}x + {c:.3f}")
```

```
Slope (m): 0.8772631578947365
Intercept (c): 0.7925263157894746
Equation: y = 0.877x + 0.793
```

1.c Calculate MSE and R² Score

```
from sklearn.metrics import mean_squared_error, r2_score

# Predict values for training data
y_pred = model.predict(X)

# Calculate MSE
mse = mean_squared_error(y, y_pred)

# Calculate R2 Score
r2 = r2_score(y, y_pred)

print("Mean Squared Error (MSE):", mse)
print("R2 Score:", r2)
```

```
Mean Squared Error (MSE): 0.07447684210526316
R2 Score: 0.9533827762426957
```

1.d Predict price for new data

```
# New unseen data
new_data = [[6]]

# Prediction
predicted_price = model.predict(new_data)

print("Predicted price for Area 6:", predicted_price[0])
```

```
Predicted price for Area 6: 6.056105263157894
```

```
c:\Users\LEN0V0\anaconda3\envs\dsaa\Lib\site-packages\sklearn\utils\
validation.py:2827: UserWarning: X does not have valid feature names,
but LinearRegression was fitted with feature names
  warnings.warn(
```

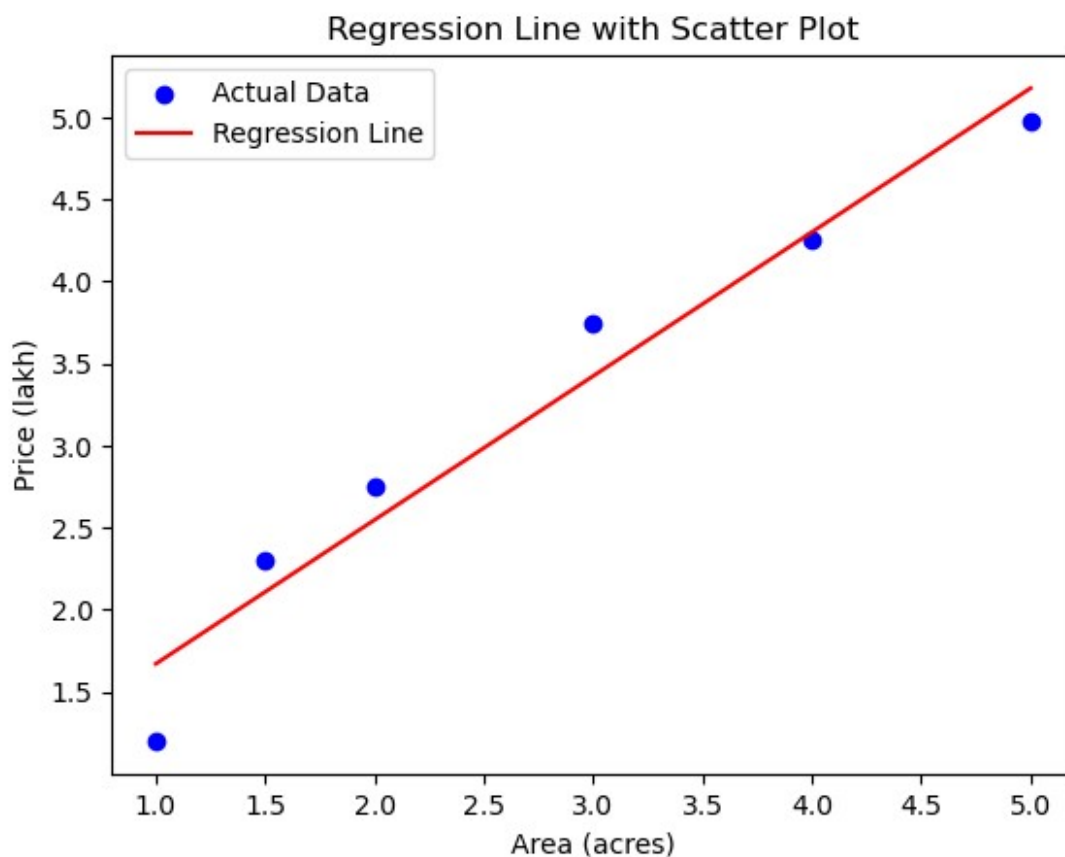

1.e Plot the regression line along with the scatter data.

```
# Plot scatter (actual data)
plt.scatter(X, y, color='blue', label='Actual Data')

# Plot regression line
plt.plot(X, y_pred, color='red', label='Regression Line')

# Labels
plt.xlabel("Area (acres)")
plt.ylabel("Price (lakh)")
plt.title("Regression Line with Scatter Plot")
plt.legend()

# Show plot
plt.show()
```



2.a Find the linear regression equation. Also determine its MSE and R² score.

```
from sklearn.preprocessing import LabelEncoder

# Dataset
data = {
    'Experience': [2, 3, 4, 5, 3],
    'Education': ['High School', 'Bachelor', 'High School',
                  'Bachelor', 'Master'],
    'Salary': [30000, 48000, 45000, 75000, 70000]
}

df = pd.DataFrame(data)

# Encode Education column
le = LabelEncoder()
df['Education_Code'] = le.fit_transform(df['Education'])

# Features and Target
X = df[['Experience', 'Education_Code']]
y = df['Salary']

# Train model
model = LinearRegression()
model.fit(X, y)

# Predictions
y_pred = model.predict(X)

# Calculate MSE
mse = mean_squared_error(y, y_pred)

# Calculate R2 Score
r2 = r2_score(y, y_pred)

print("\nMean Squared Error (MSE):", mse)
print("R2 Score:", r2)

# Display equation
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)

print("\nRegression Equation:")
print(f"Salary = {model.intercept_:.2f} + "
      f"({model.coef_[0]:.2f} × Experience) + "
      f"({model.coef_[1]:.2f} × Education_Code)")
```

Mean Squared Error (MSE): 120053333.3333334
R² Score: 0.5679047893271905
Intercept: 1100.0000000000073
Coefficients: [13566.66666667 7966.66666667]

Regression Equation:
Salary = 1100.00 + (13566.67 × Experience) + (7966.67 × Education_Code)

2.b Create a line of preference (prediction line) to see how well the regression works.

```
# Predictions
y_pred = model.predict(X)

# Plot Actual vs Predicted Salary
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.scatter(y, y_pred)

# Perfect prediction line
plt.plot([y.min(), y.max()],
         [y.min(), y.max()],
         'r--',
         linewidth=2)

plt.xlabel("Actual Salary")
plt.ylabel("Predicted Salary")
plt.title("Actual vs Predicted Salary")
plt.grid(True)

plt.show()
```

